

# Joseph Walton

Agentic AI Software Engineer | Machine Learning Engineer

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## Professional Summary

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Agentic AI and machine learning engineer with over 10 years of experience building production software, AI applications, model pipelines and cloud-based data systems across government, intellectual property and defence. Advanced Python practitioner with hands-on experience delivering agentic and generative AI services using LangGraph, LangChain, Azure AI Foundry, Azure Functions and cloud-native service-oriented architectures. Proven record of improving reliability through evaluation, observability, MLOps and zero-downtime deployment practices, while reducing cloud cost and runtime at scale. Experienced technical lead who works closely with engineers, domain experts and senior stakeholders to deliver secure, testable and commercially valuable systems.

## Technical Skills

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- **Software Engineering:** Python, SQL, Java, service-oriented architectures, CI/CD, automated evaluation, Git, Docker, Terraform
- **Agentic & Generative AI:** LangGraph, LangChain, Azure AI Foundry, Azure OpenAI, Hugging Face, RAG, vector search
- **Machine Learning:** TensorFlow, PyTorch, Scikit-learn, BERT, FastText, NLP, computer vision, recommendation systems
- **Cloud & MLOps:** Google Cloud Platform, Microsoft Azure, AWS, Vertex AI Pipelines, Azure ML Pipelines, MLflow, monitoring and observability
- **Data & Distributed Computing:** Apache Spark, PySpark, Polars, Pandas, NumPy, Cloudera
- **Databases & Search:** SQL, PostgreSQL, MySQL, Elasticsearch, HNSW vector search

## Professional Experience

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### IBM

#### Lead Data Scientist

June 2025 - Present | Contract role with central government

- Rescued underperforming AI projects by introducing engineering best practices, systematic evaluations and observability to improve production readiness and reliability.
- Designed and delivered an agentic data-processing service using Python, LangGraph, Azure AI Foundry and Azure Functions, saving case workers two hours per day and approximately £2M annually.
- Developed a Claude Skill to streamline human-in-the-loop ingestion for agent evaluation datasets, saving approximately one hour per evaluation case.
- Re-engineered a distributed Spark data-matching workload in Polars, reducing runtime from two hours to ten minutes and cloud costs by approximately £10k per month.

**Tech Stack:** Python, Polars, LangGraph, Azure AI Foundry, Azure Functions, Apache Spark

### Intellectual Property Office

#### Senior Machine Learning Engineer

November 2021 - April 2025

- Developed and operated generative AI applications within a cloud-based, service-oriented architecture using Python, LangChain, Azure OpenAI and Azure Functions.
- Built a RAG agent that helped users identify relevant trade mark Goods and Services terms from unstructured domain data.
- Implemented an end-to-end MLOps strategy across multiple environments, including evaluation and deployment workflows, reducing model release time to under one day.
- Architected resilient Polars ETL pipelines to replace Azure Databricks workloads, reducing compute costs by 50% and enabling zero-downtime deployments.

- Implemented and performance-tuned Elasticsearch HNSW vector search, balancing throughput and recall through systematic evaluation and improving search relevance by 20%.
- Conducted adversarial red-team testing of generative AI applications and translated findings into robustness improvements.
- Set the AI product roadmap and worked with technical, policy and international domain stakeholders to align systems with user and operational needs.

**Tech Stack:** Python, Azure OpenAI, Hugging Face, LangChain, Azure ML Pipelines, Azure Functions, Elasticsearch, Polars

## Incopro

### Lead Machine Learning Engineer

*February 2020 - November 2021*

- Led the design and implementation of a production MLOps framework on Google Cloud Platform, using Vertex AI Pipelines to orchestrate model training, evaluation, approval and deployment.
- Reduced time to market for client-specific model releases to under one day through reusable pipeline components and automated deployment workflows.
- Built distributed Spark pipelines on GCP to prepare large-scale text and image datasets for training, batch inference and recommendation workloads.
- Developed and deployed TensorFlow NLP and computer vision models, including a multilingual BERT model that detected intellectual property infringements across more than 90 languages.
- Built a production recommendation engine combining text and image similarity and mentored a software engineer transitioning into machine learning.

**Tech Stack:** Python, TensorFlow, Hugging Face, PySpark, Vertex AI Pipelines, Google Cloud Platform, MLflow

### Machine Learning Engineer

*November 2018 - February 2020*

- Engineered and deployed a production NLP system in a secure, containerised environment, increasing enforced infringements by 10% per analyst across approximately 200 analysts.
- Built and deployed an image-similarity model that helped secure a blue-chip client contract representing 10% of annual recurring revenue.

**Tech Stack:** Python, TensorFlow, FastText, Google Cloud Platform, MLflow, Docker

## Office for National Statistics

### Machine Learning Engineer

*September 2017 - November 2018*

- Developed Python machine learning models and distributed PySpark pipelines to securely link and process government administrative datasets at national scale.
- Contributed to a highly scalable backend for the UK Census 2021 data-collection system in a cross-functional engineering team.

**Tech Stack:** Python, PySpark, Cloudera

## Barclays

### Data Scientist

*September 2018 - October 2018*

- Seconded to develop regional economic indicators from payments data, applying statistical methods to secure financial datasets.

## Purple Secure Systems

### Software Engineer

*August 2014 - October 2016*

- Developed full-stack and big-data processing systems in Java, Python and PySpark for government and defence clients.

**Tech Stack:** Java, Python, PySpark, AngularJS

## Education

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### University of Bath

**Master of Science in Applied Mathematics**

*2016 - 2017*

**Bachelor of Science in Physics**

*2010 - 2013*

## Honors & Awards

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### Winner, ONS x Barclays Data Hackathon

Used payments data to create more granular and timely estimates for Gross Value Added (GVA).